

Technical Document – Lab 1

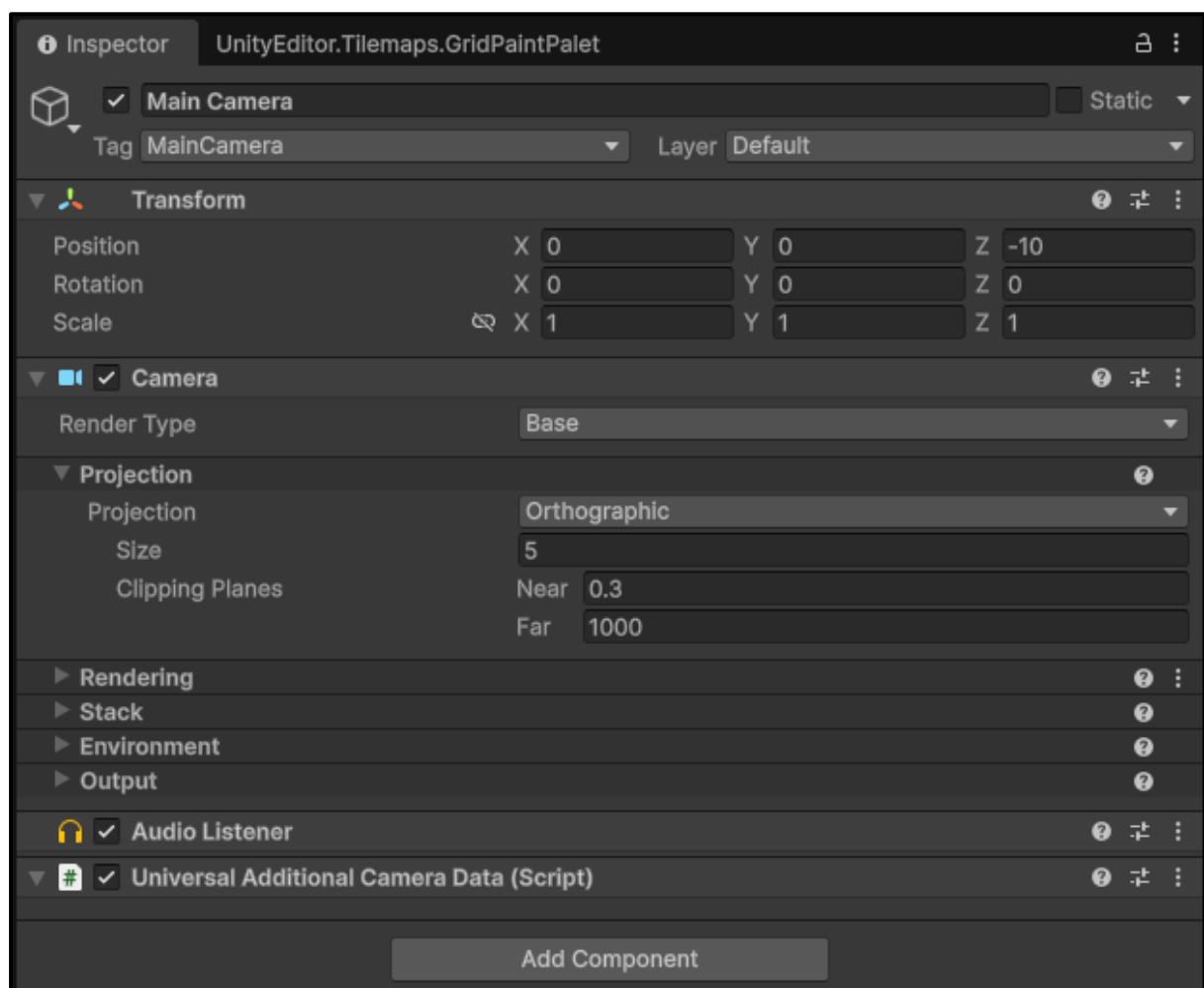
Level 1 – Game Objects and Components

A Game Object is an object used in a Unity scene. Game Objects can represent different elements of a game, such as cameras, characters, UI elements and environmental objects.

A Game Object normally contains a Transform component, which defines its position, rotation and scale in the scene.

Other components can be added to a Game Object to give it specific functionality. For example, a Camera component allows a Game Object to be used as a camera, while a Collider component allows it to interact with the physics system.

The Inspector allows the properties and components of a selected Game Object to be viewed and modified.

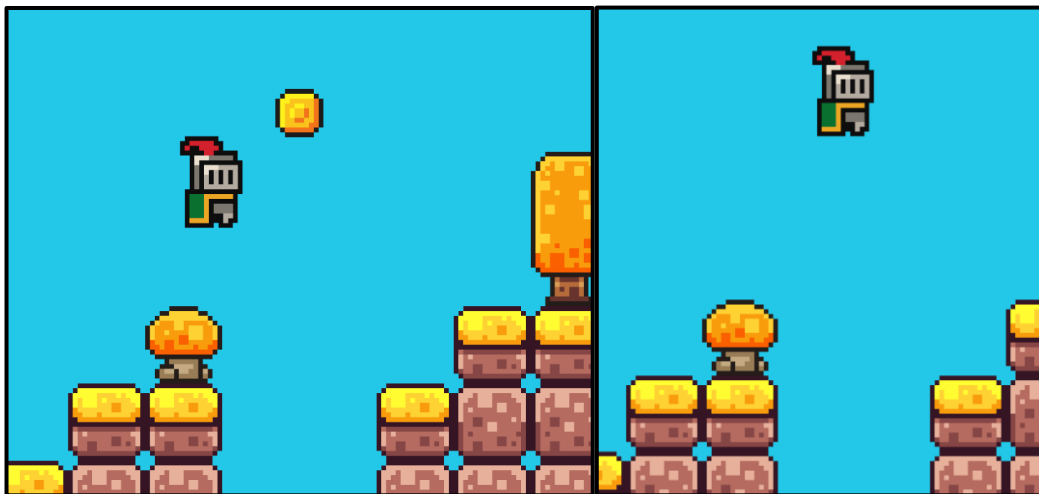


Level 2 – Collectible Behaviour

The Collectible Game Object uses the Collectible script to detect when the player interacts with it. When a Game Object with the Player tag enters its trigger collider, the script adds the defined score value to the Game Manager.

By default, the score value is 1. The collectible is also destroyed after being collected because the `destroyOnCollect` variable is enabled by default.

The Game Object therefore behaves as a collectible item that gives the player score when interacted with and is then removed from the scene.



Level 3 – Player Controller

A Player Game Object was created with a Sprite Renderer, Rigidbody2D, Box Collider2D and PlayerController script. The Player was assigned the Player tag.

A child Game Object called Ground Check was also created and positioned at the bottom of the player. It was assigned to the Ground Check field in the PlayerController component. The ground was assigned to the Ground layer, which was then selected in the Ground Layer field.

The PlayerController script handles horizontal movement and jumping. It uses the Ground Check to detect whether the player is touching the ground. The player can move horizontally and jump when grounded.

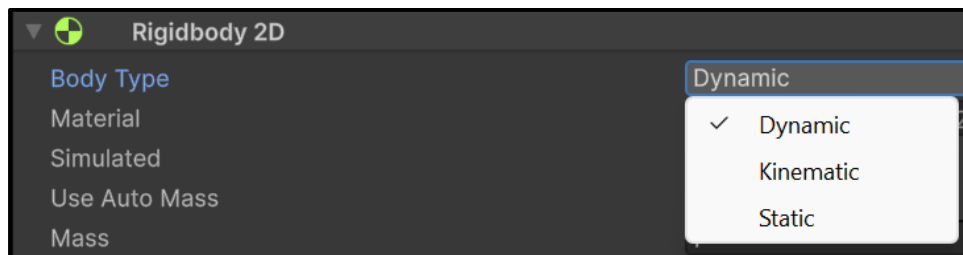
The Rigidbody2D Body Types were also tested:

Dynamic: The Game Object is affected by physics, gravity and collisions. It is suitable for a player character.

Kinematic: The Game Object is mainly controlled by its movement or scripts and is not affected by normal physics forces in the same way as a Dynamic Rigidbody2D. It could be used, for example, in moving platforms to transport the player.

Static: The Game Object is intended to remain stationary and is mainly used for objects such as walls or floors.

For the Player, the Rigidbody2D was set to Dynamic so that gravity and physics could affect the character.



Level 4 – Score and Respawn System

A Canvas was created with a TextMeshPro text object called Score, which displays the player's current score.

A Game Object called GameManager was also created and the GameManager script was attached to it. The Score TextMeshPro object was assigned to the Score Text field in the GameManager component. This allows the score displayed on screen to be updated when the player collects an item.

The Collectible script increases the player's score when the Player enters the collectible's trigger collider. The default score value of each collectible is 1.

A Respawn Trigger Game Object was created with a SimpleRespawn script and a Box Collider2D set as a trigger. The trigger was placed below the level so that it detects when the player falls.

A Respawn Point Game Object was created and positioned at the location where the player should respawn. When the player enters the Respawn Trigger, the player is returned to the Respawn Point.

Finally, the Main Menu scene was tested to verify the current game loop, including entering the level, moving the player, collecting items, updating the score and respawning after falling.

